

RAE3D: Multiscale Aggregation-Enhanced 3D Object Detection for Rail Transit Obstacle Perception

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Abstract—Point cloud-based 3D object detection technology provides precise information for detecting obstacles in front of trains. In rail transit scenarios, obstacles are usually located at a distance, and the sparsity of point cloud data leads to information loss during feature extraction and spatial transformation, adversely affecting the accuracy of obstacle detection. To tackle this issue, we propose RAE3D, a single-stage end-to-end architecture that incorporates a multiscale voxel feature aggregation module on point queries (MVA-PQ) and a bird's-eye view multilevel auxiliary (BEV-MLA) module for efficient 3D object detection. The MVA-PQ module encodes multiscale voxel features into centroids of the final voxel layer, integrating spatial and scale information. The BEV-MLA module converts shallow 3D features into sparse 2D features using height compression, guiding the network to learn the spatial structure of small objects. We also introduce a 3D centroid offset loss for optimizing bounding boxes. Extensive experiments on the Rail3D and KITTI datasets have demonstrated the superiority of the proposed RAE3D.

Index Terms—3D object detection, deep learning, rail transit, train active perception.

I. INTRODUCTION

ENSURING train operation safety is a key focus in the advancement of intelligent railway systems [1]. Train accidents caused by obstacle intrusions have become increasingly frequent, highlighting the challenges posed by the complex terrain of rail transit environments and the random, sudden, and unpredictable nature of such intrusions. The fixed-track nature of rail transportation demands precise detection of objects,

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including pedestrians and other obstacles, across long track spans. This creates stringent requirements for detection algorithms in terms of real-time performance, robustness, environmental adaptability, and scenario-specific optimization. LiDAR sensors have emerged as a preferred choice for railway obstacle detection due to their immunity to ambient lighting, resilience to adverse weather, rapid responsiveness, and accurate distance measurement. In the field of autonomous driving, LiDAR is primarily used to detect medium- and large-sized obstacles at short ranges, meeting the needs of vehicles for obstacle avoidance within relatively short braking distances. However, in the context of rail transportation, the significantly longer braking distances of trains make the detection of sparse, long-range targets a critical requirement. Addressing this gap by developing robust, high-performance algorithms for railway-specific needs is an urgent priority [2].

Point cloud obstacle detection is a vital technology for enhancing safety and reliability in railway transportation, addressing the unique challenges of obstacle detection in complex rail environments [3]. By providing precise 3D spatial information, it enables systems to accurately identify personnel, vehicles, and other obstacles on the track in real time, helping to prevent collisions and ensure overall train safety [4]. Unlike traditional 2D object detection methods, 3D object detection addresses the limitations associated with ignoring depth information and spatial structure. With the widespread adoption of sensors such as LiDAR and RGB-D cameras, capturing detailed 3D scene information has become increasingly feasible, positioning 3D object detection at the forefront of research in intelligent railway systems. Existing LiDAR-based 3D object detection methods for railways primarily rely on background subtraction and clustering techniques [5]. With the increasing application of deep learning in point cloud object detection, several deep learning-based approaches have been developed and explored, such as SECOND [6], PointPillars [7], Point R-CNN [8], 3DSSD [9], and PV R-CNN [10]. These deep learning methods offer robust learning capabilities and high recognition accuracy. Despite extensive research on LiDAR-based recognition methods, challenges related to accuracy and speed persist when these methods are scaled for railway applications.

In rail transit scenarios, obstacles are typically sparse and often located at considerable distances. 3D sparse convolution

provides an efficient sparse representation and multiscale feature encoding of voxelized point clouds, enabling rapid downsampling while preserving critical geometric information. The local set abstraction (SA) method based on PointNet++ [11] uses flexible spherical receptive fields to precisely capture local geometric details of point clouds, mimicking the focus mechanism of the human eye when observing complex scenes. This approach is particularly effective for characterizing small obstacles with intricate geometries. Integrating the strengths of 3D sparse convolution and local SA enhances the detection performance for obstacles over long-distance rail tracks. To address this, we propose RAE3D, a single-stage end-to-end architecture that completes the entire 3D object detection process, including feature extraction, feature aggregation, and bounding box prediction, in a single forward pass of the network. It provides high inference efficiency and real-time performance, rendering it highly suitable for rail transportation scenarios. The architecture incorporates a multiscale voxel feature aggregation module based on point query (MVA-PQ) and a bird's-eye view multilevel auxiliary (BEV-MLA) module. In addition, we introduce a 3D center offset loss to refine the predicted bounding box centers, ensuring better alignment with actual geometric centers. This improvement enhances the robustness of the model in detecting parallel or distant obstacles. The contributions of this work are summarized as follows.

- 1) We design the MVA-PQ module to integrate the global representation capability of 3D sparse convolution with the local precision of SA. This design effectively addresses the varying scale requirements of object detection in rail transit by aggregating multiscale information through pointwise queries.
- 2) To address the uneven distribution of nonempty voxels in rail transit scenarios, the BEV-MLA module is designed to enhance the fusion of multiscale spatial semantic features. It maps 3D sparse voxel features onto multiscale BEV space and employs sparse height compression along with 2D sparse convolution.
- 3) By introducing a 3D centroid offset loss function, we optimize the center positions of predicted bounding boxes, aligning them more closely with the geometric centers of the actual objects.
- 4) Extensive empirical studies and ablation analyses conducted on the KITTI [12] and custom Rail3D datasets demonstrate that our proposed RAE3D method exhibits significant improvements in both accuracy and efficiency, validating its effectiveness and generalization capability.

The rest of this article is organized as follows. Section II reviews related works, Section III introduces the proposed RAE3D method, and Section IV presents the experimental analysis. Finally, Section V concludes this article.

II. RELATED WORK

In this section, we briefly review existing research on 3D object detection and its related studies in the field of rail transportation.

A. LIDAR-Based 3D Object Detection

Voxel-based, point-based, and hybrid voxel-point methods have introduced innovative contributions, significantly advancing the field of 3D object detection. Voxel-based 3D object detection, inspired by deep learning techniques for 2D image processing, transforms irregular point clouds into regular 3D grids, utilizing 3D or 2D convolutions to generate structured 4D feature maps. VoxelNet [13] enhances 3D object representation by replacing handcrafted features with voxel feature encoding (VFE), significantly improving detection performance in complex scenes. SECOND [6] further optimizes this approach by using sparse convolutions, enabling faster training. PointPillars [7] and PillarNet [14] introduce vertical columns (pillars), eliminating the need for 3D convolutions and integrating with 2D detection architectures. Attention-based grid architectures [15], [16], leveraging Transformers, replace traditional convolutions, enhancing voxel structure perception and global context extraction. Despite addressing irregular point cloud structures, grid-based methods are prone to quantization errors, leading to information loss and higher computational costs. SIEV-Net [17] tackles this by introducing a hierarchical VFE module and a height information complement module, exploiting the uneven spatial distribution and height of LiDAR point clouds to improve detection. PDV [18] addresses LiDAR point density variations by integrating density-aware voxel features, RoI grid pooling with KDE and self-attention, and point density-based bounding box refinement. OcTr [19] adopts an octree-based Transformer, capturing rich global context with a coarse-to-fine approach and enhancing foreground perception through hybrid positional embedding. MsSVT++ [20] proposes a mixed-scale sparse voxel transformer, capturing both long-range and fine-grained information and improving localization through a center voting module, while addressing computational complexity with chessboard sampling.

Point-based methods directly detect objects in point cloud scenes, using SA to aggregate features for bounding box prediction and classification. VoteNet [21] employs hough voting for object center prediction, while 3DSSD [9] balances accuracy and speed by retaining relevant points. PointRCNN [8] generates high-quality proposals directly from raw point clouds in a bottom-up manner and refines them using canonical coordinates. CenterPoint [22] innovatively detects object centers using a key-point detector, regresses attributes such as 3D size, orientation, and velocity. Although these methods retain geometric details and achieve high accuracy, optimizing sampling, aggregation, and computational efficiency remains challenging.

Hybrid models effectively combine the strengths of voxel-based and point-based methods to overcome their respective limitations in 3D object detection. For example, SA-SSD [23] enhances localization precision by leveraging 3D point cloud structure through an auxiliary network, introducing point-level supervision and part-sensitive warping, which improves detection without increasing inference costs. PV-RCNN [10] incorporates a voxel SA module and RoI-grid pooling to improve computational efficiency while enriching context

encoding. PVT-SSD [24] innovatively integrates voxel-based sparse convolutions with an input-dependent query initialization module and a virtual range image module, which enhances feature encoding and localization accuracy, effectively combining point and voxel representations. In addition, HEDNet [25] employs hierarchical encoder-decoder blocks to capture long-range dependencies, improving the detection of large and distant objects in sparse 3D point clouds with both accuracy and efficiency. PIPC-3Ddet [26] introduces a perspective information embedding module and a proposal correlation reasoning module to refine voxel features and 3D proposals, further boosting detection performance. Lastly, a domain-adaptive subterranean 3D pedestrian detection method [27] adapts pretrained detectors from road scenes to unannotated subterranean environments using an instance transfer-based strategy and a pseudolabel confidence-guided learning mechanism, significantly improving performance in low-illumination conditions. This work aims to further explore and exploit hybrid models for 3D object detection, particularly in rail transportation applications.

B. 3D Object Detection in Railway

3D object detection has become a key technology for intelligent perception and scene understanding in railway transportation [28]. Recent research has explored various methods to enhance railway obstacle detection. Wisultschew et al. [29] presented an IoT-based object detection system, demonstrating real-time performance and collision avoidance capabilities through the use of an embedded edge platform and LiDAR sensors in railway level crossing scenarios. Qu et al. [30] proposed an obstacle detection method based on LiDAR, which preprocesses point clouds and fits rails using the RANSAC algorithm, incorporating an adaptive distance threshold to improve segmentation accuracy in sparse point clouds. Nan et al. [31] proposed a high-precision obstacle detection algorithm using 3D mechanical LiDAR, demonstrating enhanced detection accuracy and stability through dynamic threshold modulation and local feature analysis using PCA and local-ICP clustering techniques. Nan et al. [32] presented an advanced LIDAR-based obstacle monitoring system, demonstrating significant potential in enhancing railway safety through precise point cloud transformation and robust detection algorithms. Zhu et al. [33] introduced a railway surface obstacle extraction algorithm combining global planar feature-based coarse registration with iterative closest point (ICP) registration.

While 3D object detection has been extensively studied in autonomous vehicles, research in railway transportation has largely focused on obstacle detection and track recognition in static, linear scenarios, with forward obstacle detection based on onboard sensors remaining significantly underexplored. To bridge this gap, this article conducts an in-depth investigation into forward clearance 3D obstacle detection for trains, leveraging onboard LiDAR sensors to enhance the autonomous safety capabilities of trains in dynamic rail environments.

III. METHODOLOGY

A. Framework Overview

In this section, we introduce the design of the proposed RAE3D, a single-stage 3D object detection framework. As shown in Fig. 1, the overall architecture of RAE3D primarily consists of the following components. 1) Input: Voxelization of the input point cloud. 2) 3D backbone: Extracting high-level features from VFE-encoded voxels using sparse convolutions, with the MVA-PQ module designed to extract and aggregate multiscale voxel features. 3) 2D backbone: Employing sparse height compression (SHC) to generate BEV feature maps, with a BEV-MLA module to further extract and fuse 2D features. 4) Detection head: Responsible for object classification, bounding box regression, and IoU prediction. The following sections provide a detailed explanation of these modules.

B. MVA-PQ Module

Existing voxel-based methods often suffer from quantization loss during voxelization. Therefore, we propose the MVA-PQ module into the 3D Voxel CNN to aggregate multiscale voxel features at the final down-sampling layer. As shown in Fig. 2, we introduce the grid centers of the feature volumes at this layer as query points. By querying within predefined spherical regions, we aggregate shallowly encoded voxel features, enabling multiscale spatial encoding from low-level to high-level layers.

1) *Voxel Feature Encoding*: Initially, spatial points are converted into voxel cell with a spatial resolution of $L \times W \times H$, as described in (1). The point cloud information within each voxel cell (i.e., 3D coordinates (x, y, z) and reflectance intensity r) is averaged to obtain the initial encoded voxel features $F^{(l_0)} = \{f_1^{(l_0)}, \dots, f_{N_0}^{(l_0)}\}$, with the corresponding spatial coordinates represented as $V^{(l_0)} = \{v_1^{(l_0)}, \dots, v_{N_0}^{(l_0)}\}$, where N_0 is the number of nonempty voxels. As shown in Fig. 2(a), sparse convolution in the 3D backbone network enables hierarchical down-sampling and multiscale feature extraction, as described in (2), encoding the features of nonempty voxels into 3D sparse feature volumes at down-sampling scales of $1 \times, 2 \times, 4 \times$, and $8 \times$

$$\Phi_v(v_j) = \{p_i \in P | \Phi_p(p_i)\}, v_j \in V \quad (1)$$

$$f_j^{(l+1)} = \sum_{v_k^l \in K(v_j^{l+1})} W \cdot f_k^l + b \quad (2)$$

where $\Phi_p(\cdot)$ maps a point p_i to the voxel v_j it belongs to, while $\Phi_v(\cdot)$ maps a voxel to the corresponding point set. P is the set of spatial points, and V is the set of nonempty voxel grids. $K(v_j^{l+1})$ represents the set of voxel positions forming the neighborhood of the convolution kernel.

2) *Multiscale Voxel Feature Aggregation Module*: We propose the MVA-PQ module, which encodes multiscale voxel features into the center points of the 3D CNN output feature volume, thereby enhancing the variable receptive field. As illustrated in

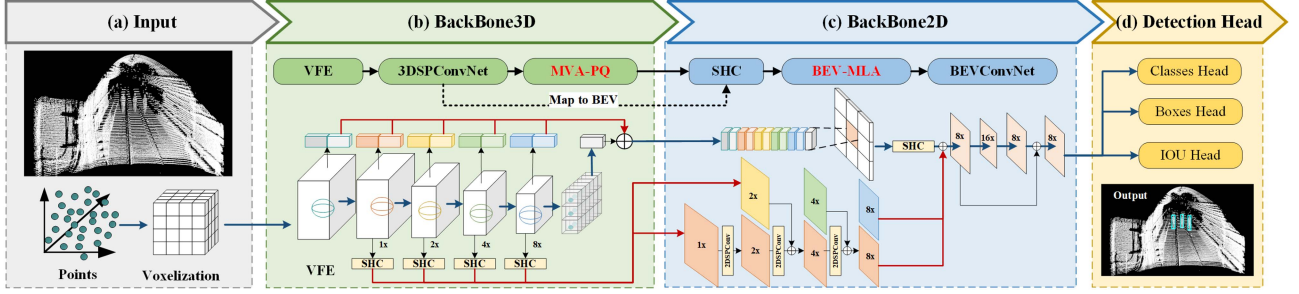


Fig. 1. Overview of RAE3D architecture.

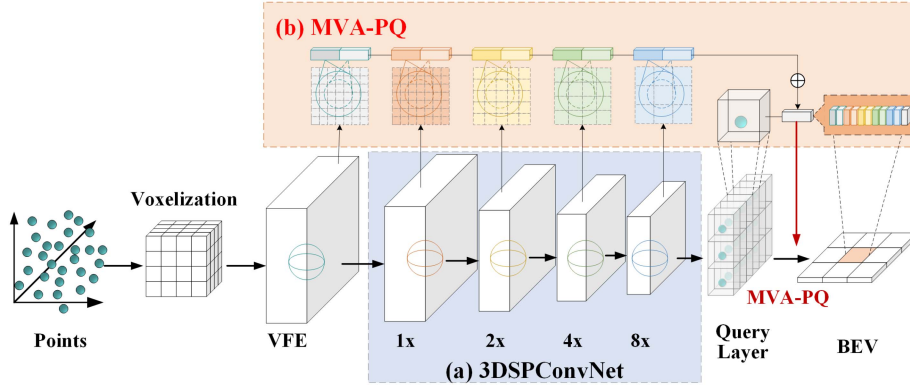


Fig. 2. Schematic diagram of MVA-PQ.

Fig. 2(b), we employ SA operations to sample the neighborhood around each center point. MLP and pooling operations are then applied within each independent grouping region to aggregate the features. By setting different spherical radii, local multiscale features of the center points are obtained. Hierarchical sampling and aggregation operations are then performed to summarize multiscale geometric features. The specific operations are as follows.

We designate the final down-sampling layer of the 3D Voxel CNN as the aggregation layer, with the voxel cell centers in this layer serving as the query points Q for SA. The spatial coordinates of the query points $Q = \{q_1, \dots, q_m\}$ are calculated based on the voxel indices and actual voxel dimensions, where m represents the number of query points $\frac{L}{8} \times \frac{W}{8} \times \frac{H}{8}$. The set of voxel feature vectors for this layer is denoted as $F^{(q)} = \{f_1^{(q)}, \dots, f_m^{(q)}\}$. The spatial coordinates of the voxels in the k th down-sampling layer are denoted as $V^{(l_k)} = \{v_1^{(l_k)}, \dots, v_{N_k}^{(l_k)}\}$, and the corresponding set of voxel feature vectors is $F^{(l_k)} = \{f_1^{(l_k)}, \dots, f_{N_k}^{(l_k)}\}$, where N_k is the number of nonempty voxels in the k th layer. It is important to note that query points may exist in empty voxels, while the multiscale voxel cells being queried are all nonempty. This approach facilitates the propagation of features from dense regions to sparse regions.

For each query point q_i , grouped sampling is performed on the nonempty voxels of the k th layer within the radius neighborhood r_k , to obtain the feature vector set of the grouped voxels, which

can be expressed as

$$\mathcal{G}_i^{(l_k)} = \left\{ \mathcal{S}_{\eta=16}[v_j^{(l_k)} - q_i; f_j^{(l_k)}] \mid \begin{array}{l} \|v_j^{(l_k)} - q_i\|^2 < r_k \\ j = 1, \dots, N_k \end{array} \right\} \quad (3)$$

where $\mathcal{S}_{\eta=16}[\cdot]$ represents the process of collecting the 16 nearest voxels within the neighborhood r_k , and concatenating their local relative coordinates $v_j^{(l_k)} - q_i$ with the features $f_j^{(l_k)}$.

Subsequently, the PointNet module is used to generate the local aggregated features for the query point

$$f_i^{(qv_k)} = \text{MAX}(\mathcal{A}(\mathcal{G}_i^{(l_k)})) \quad (4)$$

where $\mathcal{A}(\cdot)$ denotes a three-layer MLP network that encodes the features. The operation $\text{MAX}(\cdot)$ represents max pooling along the channels, which aggregates the multiscale features into the query point. In addition, two different scales r_k are used in the k th layer to increase the local variable receptive field.

The voxel features from different layers of the 3D CNN are concatenated and aggregated into the multiscale voxel features q_i for the query point $f_i^{(qv)}$

$$f_i^{(qv)} = \{f_i^{(q)} \oplus f_i^{(qv_k)} \mid k = 0, 1, 2, 3, 4\} \quad (5)$$

C. BEV-MLA Module

Inspired by the feature pyramid network (FPN) used in 2D object detection, we design the BEV-MLA module during the

Algorithm 1: Training Procedure of the Proposed Method.

Input: N point cloud data ($P_{[N,4]}$) with 3D coordinates x, y, z and reflectance r , voxel space shape $S_{L \times W \times H}$, loss hyperparameters ($\lambda_1, \lambda_2, \lambda_3, \lambda_4$), number of anchors (n_a), number of categories (n_c)

Output: training loss (L)

- 1: batch(B) $\leftarrow 4$, epochs $\leftarrow 80$
- 2: **for** epoch **in** range(epochs):
- 3: **(1) Initialization:**
- 4: **Voxelization**($P_{[N,4]}$)
- 5: $\rightarrow N_{[N,4]}$ (4:batch index and voxel center coordinates X, Y, Z)
- 6: **(2) 3D Backbone:**
- 7: **MeanVFE**($P_{[N,4]}, N_{[N,4]}$)
- 8: $\rightarrow F_{[N_0,4]}^{l_0}, V_{[N_0,4]}^{l_0}$ (4:x,y,z,r N_0 :number of nonempty voxels)
- 9: **3DSPConvNet**($F_{[N_0,4]}^{l_0}, N_{[N_0,4]}^{l_0}$)
- 10: $\rightarrow T_{Conv0} : (F_{[N_0,4]}^{l_0}, V_{[N_0,4]}^{l_0}, S_{L \times W \times H})$
- 11: $\rightarrow T_{Conv1} : (F_{[N_1,C_1]}^{l_1}, V_{[N_1,4]}^{l_1}, S_{L \times W \times H})$
- 12: $\rightarrow T_{Conv2} : (F_{[N_2,C_2]}^{l_2}, V_{[N_2,4]}^{l_2}, S_{L/2 \times W/2 \times H/2})$
- 13: $\rightarrow T_{Conv3} : (F_{[N_3,C_3]}^{l_3}, V_{[N_3,4]}^{l_3}, S_{L/4 \times W/4 \times H/4})$
- 14: $\rightarrow T_{Conv4} : (F_{[N_4,C_4]}^{l_4}, V_{[N_4,4]}^{l_4}, S_{L/8 \times W/8 \times H/8})$
- 15: $\rightarrow T_q : (F_{[N_4,C_5]}^q, V_{[N_4,4]}^q, S_{L/8 \times W/8 \times H/8})$
- 16: **MVA-PQ**
($T_{Conv0}, T_{Conv1}, T_{Conv2}, T_{Conv3}, T_{Conv4}, T_q$)
- 17: $\rightarrow T_{qv} : (F_{[N_4,C_6]}^{qv}, V_{[N_4,4]}^q, S_{L/8 \times W/8 \times H/8})$
- 18: **(3) 2D Backbone:**
- 19: **SHC**($T_{qv}, T_{Conv1}, T_{Conv2}, T_{Conv3}, T_{Conv4}$)
- 20: $\rightarrow (F_{[B,C_6,L/8,W/8]}^{qv}, F_{[B,C_1,L,W]}^{Conv1}, F_{[B,C_2,L/2,W/2]}^{Conv2}, F_{[B,C_3,L/4,W/4]}^{Conv3}, F_{[B,C_4,L/8,W/8]}^{Conv4})$
- 21: **BEV-MLA**($F^{qv}, F^{Conv1}, F^{Conv2}, F^{Conv3}, F^{Conv4}$)
- 22: $\rightarrow F_{[B,C_7,L/8,W/8]}^{bev}$
- 23: **BEVConvNet**(F^{bev}) $\rightarrow F_{[B,C_7,L/8,W/8]}^{2D}$
- 24: **(4) Detection Head:**
- 25: **2DConvNet**(F^{2D})
- 26: $\rightarrow (cls_{[B,L/8,W/8,n_a \times n_c]}, loc_{[B,L/8,W/8,7n_a]}, dir_{[B,L/8,W/8,2n_a]}, cen_{[B,L/8,W/8,3n_a]})$
- 27: **Return:** $L = \frac{1}{N_{pos}} (\lambda_1 L_{cls} + \lambda_2 L_{loc} + \lambda_3 L_{dir} + \lambda_4 L_{cen})$

height compression process, as shown in Fig. 3. This module maps 3D voxel sparse features to multiscale BEV spaces through fully sparse height compression operations and progressively fuses multilevel spatial geometric information using 2D sparse convolutions. This approach efficiently captures long-range sparse target features. Finally, the fused multiscale 2D sparse features are fed into a 2D backbone network for dense feature extraction and 3D object prediction.

1) **BEV Multilevel Auxiliary Branch:** Existing methods [6], [10], [16] achieve height compression by combining the Z-dimension with the channel dimension, compressing 3D voxel features into dense BEV mappings. However, these methods are memory-intensive. To address this, we adopt the SHC method

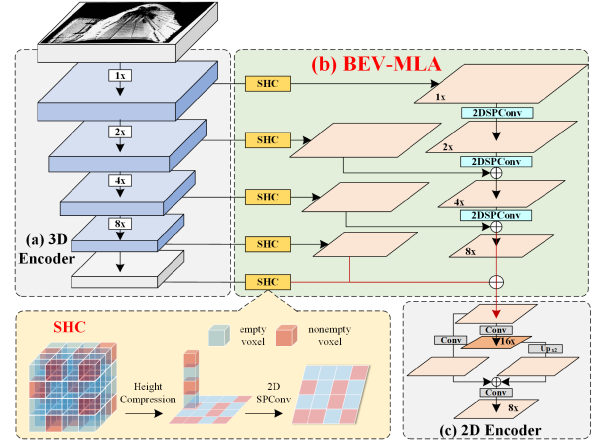


Fig. 3. Schematic diagram of BEV Multi-Level Auxiliary Module.

from [36], which maps nonempty voxels onto the XY plane and accumulates features at identical locations using (6). This approach maintains sparsity, reducing computational overhead. In addition, we introduce a BEV multilevel auxiliary branch in the neck module. Using (7), down-sampling is applied to the SHC features from the four layers of the 3D CNN (with down-sampling rates of $1 \times$, $2 \times$, $4 \times$, and $8 \times$). These features are fused across layers using (8), enhancing the receptive field and feature representation. As shown in Fig. 3(b), this method efficiently processes sparse data while preserving critical height information for BEV extraction

$$F_{SHC}^l = \left\{ f_s | f_s = \sum_{v_i \in Z(v_s)} f_i \right\}, l \in \{1, 2, 3, 4\} \quad (6)$$

$$F_{DS}^{l+1} = \text{DownSample}(F_{SHC}^l), l \in \{1, 2, 3\} \quad (7)$$

$$F_{Fusion}^{l+1} = F_{DS}^l \oplus F_{SHC}^l, l \in \{1, 2, 3, 4\} \quad (8)$$

where F_{SHC} denotes the features after sparse height compression, with $Z(\cdot)$ as the set of nonempty voxels at 2D position v_s and f_s as the feature value at this location. F_{DS} and F_{Fusion} represent the down-sampled features and the fused features at the current layer, respectively.

2) **2D Backbone Network:** The fused multiscale 2D sparse features are transformed into dense feature representations and fed into a 2D backbone network for further processing of BEV features and generating the final detection output. As shown in Fig. 3(c), this network consists of two subnetworks: a top-down subnetwork that performs 1×1 convolutions and 3×3 down-sampling operations on high-density 2D features (resulting in $16 \times$ down-sampled features), and an up-sampling subnetwork that restores the spatial resolution to an $8 \times$ scale. Finally, features from all layers are concatenated with the original input features and passed to the next stage for 3D object detection. This process is mathematically described as follows:

$$F_{2D} = \text{Conv}_{1 \times 1} \{ \text{Conv}_{1 \times 1}(F_{8 \times}) \oplus \text{Up}_{\times 2}(\text{Conv}_{3 \times 3}(F_{8 \times})) \} \\ |F_{8 \times} = \text{Dense}(F_{fusion}) \quad (9)$$

where Conv represents the convolutional feature extraction layer, $\text{Up}_{\times 2}$ denotes the up-sampling layer, and Dense converts sparse features into dense representations.

D. Loss Function

We adopt the same multitask loss function as in [6] to jointly optimize the RAE3D architecture, which includes classification loss L_{cls} , localization loss L_{loc} , and orientation loss L_{dir} . In addition, we introduce a centroid offset loss L_{cen} for 3D regression boxes to directly optimize the center position of the predicted box, ensuring that the centroid of the predicted box is closer to that of the ground truth box. Specifically, we perform clustering analysis on the point cloud data within the bounding boxes, calculate the offset between the centroids of the predicted and ground truth boxes, and use this offset to guide model training. The calculation of L_{cen} is as follows:

$$L_{\text{cen}} = \frac{\|C_{\text{box}} - C_{\text{gt}}\|_2}{\eta^2} \quad (10)$$

$$C_{\text{box}} = \frac{\sum_{i=1}^{N_b} p_i K\left(\frac{C_{\text{box}} - p_i}{\gamma}\right)}{\sum_{i=1}^{N_b} K\left(\frac{C_{\text{box}} - p_i}{\gamma}\right)} \quad (11)$$

$$C_{\text{gt}} = \frac{\sum_{i=1}^{N_g} p_j K\left(\frac{C_{\text{gt}} - p_j}{\tau}\right)}{\sum_{i=1}^{N_g} K\left(\frac{C_{\text{gt}} - p_j}{\tau}\right)} \quad (12)$$

where C_{box} and C_{gt} represent the centroids of the predicted and ground truth bounding boxes, respectively. The L2 norm loss $\|\cdot\|_2$ between them is computed and normalized by the size of the bounding boxes using the normalization factor η . For each bounding box, we compute the centroid of N points within the box using Kernel density estimation (KDE). The weight of each point is calculated using the Gaussian kernel function $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$, where the bandwidth parameters γ and τ are derived from the standard deviation and mean of the point cloud coordinates. KDE captures the spatial or feature distribution of the input data, and the choice of the Gaussian kernel ensures smoothness and robustness in parameter estimation, dynamically adjusting according to the data density.

The overall loss function is defined as follows:

$$L = \frac{1}{N_{\text{pos}}} (\lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{loc}} + \lambda_3 L_{\text{dir}} + \lambda_4 L_{\text{cen}}) \quad (13)$$

where N_{pos} represents the number of positive anchors $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are hyperparameters used to balance the weights of each component of the loss.

IV. EXPERIMENTS

In this section, we evaluate the performance of our model using the KITTI benchmark dataset and our custom Rail3D dataset, which features rail transit scenarios.

A. Dataset and Implementation Details

1) **Datasets:** The KITTI [12] dataset is created by the Karlsruhe Institute of Technology in Germany and the Toyota Technological Institute at Chicago. It consists of image data collected

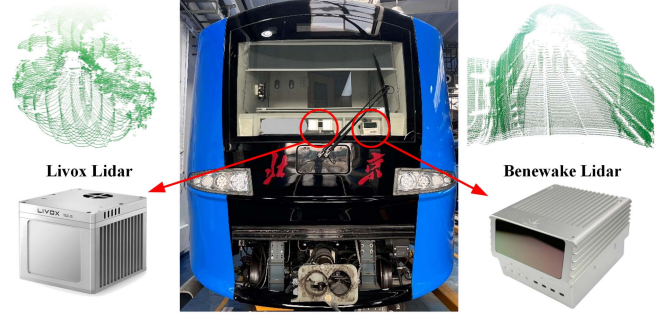


Fig. 4. Experimental platform.

from a variety of real-world scenes, including both urban and rural environments, making it one of the most widely used benchmarks for 3D object detection in autonomous driving. The dataset contains 7481 training samples and 7518 testing samples. The training set is further divided into a train split with 3712 examples and a validation split with 3769 examples.

In addition, we develop a custom 3D dataset for rail transit scenarios, named Rail3D. The point cloud data for this dataset was captured using a LiDAR system installed at the front of the train, as shown in Fig. 4. We use two different LiDAR devices to collect data, resulting in significantly different point cloud distributions, which are employed to verify the robustness of the method presented in this article. This dataset encompasses both subway tunnel and open railway scenes, simulating scenarios where people and obstacles, such as cardboard boxes and foam rocks, encroach upon the tracks. Rail3D contains a total of 3058 frames with annotations for two categories: “Obstacle” and “Pedestrian,” including 6703 pedestrian samples and 3190 obstacle samples. The dataset is divided into 2140 frames for training and 918 frames for validation.

2) **Implementation Details:** For the KITTI dataset, all parameters are set to default. For the Rail3D dataset, we configure the spatial range limits of the point cloud as follows: x from 0 to 360 m, y from -40 to 40 m, and z from -8 to 8 m. The resolution along the x -, y -, and z -axes is set to 0.1, 0.1, and 0.4 m, respectively. We allow a maximum of 16 000 voxels, with each voxel containing up to 20 points. We adopt the same anchors and matching strategy as in SECOND, where each anchor is defined by its width, length, height, and center point, and is applied in two orientations: 0° s and 90° . For the “Obstacle” and “Pedestrian” classes, the anchor dimensions (width, length, height) are set to (0.7, 0.7, 0.7) m and (0.7, 0.7, 1.8) m, respectively. The positive and negative matching thresholds are set to 0.8 and 0.6.

Our model is trained on an NVIDIA GeForce RTX 3090 GPU using the ADAM optimizer. The initial learning rate is set to $1e-3$, with a batch size of 4, across a total of 80 epochs. A cosine annealing strategy is used to decay the learning rate over time, with weight decay and momentum factors set to 0.01 and 0.9, respectively.

3) **Evaluation Metrics:** For the KITTI dataset, the performance of 3D object detection algorithms is evaluated using AP-R40 [37], which approximates the shape of the Precision/Recall curve, as shown in (14). The evaluation metrics include average

TABLE I

PERFORMANCE COMPARISON ON THE RAIL3D VALIDATION SET AND THE LATENCY IS MEASURED BY US BASED ON THE SAME ENVIRONMENT (I.E., RTX 3090 GPU) WITH FOUR BATCH SIZE

Method	3D mAP	Obs.		Ped.		Latency (ms)
		3D	BEV	3D	BEV	
SECOND [6]	72.86	57.79	80.91	87.93	91.72	33.08
PointPillars [7]	58.42	43.11	74.47	73.73	89.86	19.03
Part-A2 [34]	82.87	74.08	88.19	91.65	92.38	74.91
VoxelR-CNN [35]	81.21	73.76	87.76	88.66	89.2	43.09
CenterPoint [22]	79.23	67.26	86.56	91.19	94.43	89.00
PillarNet [14]	69.78	58.75	82.46	80.8	92.16	62.50
PointRCNN [8]	74.12	69.11	80.82	79.12	81.54	87.21
PV-RCNN [10]	84.93	76.28	87.94	93.57	96.16	105.67
PDV [18]	83.83	75.53	86.61	92.13	94.97	75.88
Ours	87.51	79.36	89.99	95.65	98.16	23.52
Improvement	+2.58	+3.08	+2.05	+2.08	+2.00	

precision (AP) across three difficulty levels (easy, moderate, and hard) and its variants, such as BEV AP (based on bird's-eye view detections) and 3D AP (based on full 3D bounding boxes). Intersection over union (IoU) thresholds of 0.7, 0.5, and 0.5 are applied for vehicles, pedestrians, and cyclists, respectively. In addition, the 3D mean average precision (3D mAP) across the three difficulty levels is computed to assess the overall performance of detection methods for each category. For the custom Rail3D dataset, we employ the same evaluation metrics as those used for the KITTI dataset

$$AP|_{R_{40}} = \frac{1}{R_{40}} \sum_{r \in R_{40}} \rho_{\text{interp}}(r) \quad (14)$$

where $R_{40} = \{1/40, 2/40, 3/40, \dots, 1\}$ averaging precision results on 40 recall positions. The interpolation function is defined as $\rho_{\text{interp}}(r) = \max_{r' \geq r} \rho(r')$, where $\rho(r)$ gives the precision at recall r .

B. Evaluation Results on Rail3D and KITTI Dataset

1) Rail3D Dataset Quantitative Analysis: We utilize the official KITTI evaluation metric, AP-R40, to assess the detection performance of our proposed method on the Rail3D dataset. Since the custom dataset does not include predefined difficulty levels, the metrics in Table I are refined to present both 3D and BEV average precision. As presented in Table I, the proposed RAE3D method (denoted as “Ours”) achieves significant improvements in 3D detection accuracy and BEV accuracy, highlighting its superiority in precision performance. Specifically, RAE3D demonstrates excellent real-time performance in rail transit object detection, achieving a latency of 23.52 ms, which corresponds to over 42 frames per second (FPS) on an RTX 3090 GPU. Compared to other methods, it outperforms state-of-the-art approaches like PDV (75.88 ms), PV-RCNN (105.67 ms), and Voxel R-CNN (43.09 ms) in terms of efficiency, while maintaining the highest 3D mAP of 87.51%. Although PointPillars achieves slightly lower latency (19.03 ms), its accuracy (58.42%) is significantly lower. The proposed method effectively balances accuracy and speed, making it ideal for real-time rail transit applications.

To intuitively demonstrate the effectiveness of the proposed RAE3D, Fig. 5 presents qualitative results from several examples in the test set. As shown in the first set of visualizations, RAE3D exhibits stronger detection capabilities for distant pedestrians, with lower false positive and missed detection rates, as well as higher confidence in obstacle localization. In contrast, PV-RCNN and PDV display more missed detections or prediction errors, as indicated by the red and cyan circles in the figure.

2) KITTI Dataset Quantitative Analysis: In Table II, we compare the performance of the proposed RAE3D with other state-of-the-art 3D detectors on the KITTI validation set. The comparison includes models based on voxels, point clouds, and voxel-point cloud approaches. The evaluation metrics cover the mean precision values (AP-R40) for three target categories—vehicles, pedestrians, and cyclists—across easy, moderate, and hard difficulty levels, as well as the overall mean precision (mAP).

Table II shows that the proposed RAE3D method achieves superior performance on the KITTI validation set, with an overall mAP of 76.30%, closely approaching the best-performing method, PIPC-3Ddet (77.81%), and outperforming other two-stage methods such as MsSVT++ (74.32%) and OcTr (70.79%). Leveraging its framework designed for distant and small objects, RAE3D demonstrates outstanding pedestrian detection accuracy, achieving 66.40% (moderate) and 60.07% (hard), comparable to SIEV-Net (66.52%, 61.16%) and surpassing MsSVT++ (59.58%, 53.92%). For cyclist detection, RAE3D achieves mAP scores of 75.32% (moderate) and 70.58% (hard), outperforming PIPC-3Ddet (72.59%, 69.81%). Moreover, RAE3D maintains competitive efficiency with a latency of 35 ms, second only to PointPillars (24 ms), faster than one-stage methods such as SIEV-Net (45 ms) and comparable to 3DSSD (39 ms). This demonstrates the ability of RAE3D to balance high detection accuracy and low computational cost, making it suitable for real-time 3D object detection applications. Qualitative results on the KITTI validation set are presented in Fig. 6, highlighting the advantage of RAE3D in detecting distant, small, and occluded objects, with significantly reduced missed detection rates. Furthermore, RAE3D successfully identifies unmarked objects that are either too distant or heavily occluded, as indicated by the green circles in the figure.

C. Ablation Study

1) Analysis of RAE3D Components Improvements: To validate the effectiveness of the key components in RAE3D, we conduct ablation experiments on the validation subsets of the KITTI and Rail3D datasets by comparing the MVA-PQ module, BEV-MLA module, and center offset loss component with the baseline method (SECOND_IoU). The ablation results, presented in Table III, highlight the performance improvements contributed by each component, with the gains marked in red.

As shown in the Table III, all three components significantly enhance the detection performance. Specifically, compared to the baseline method, the addition of the MVA-PQ module improves 3D mAP by 3.36% and 5.47% on the KITTI and

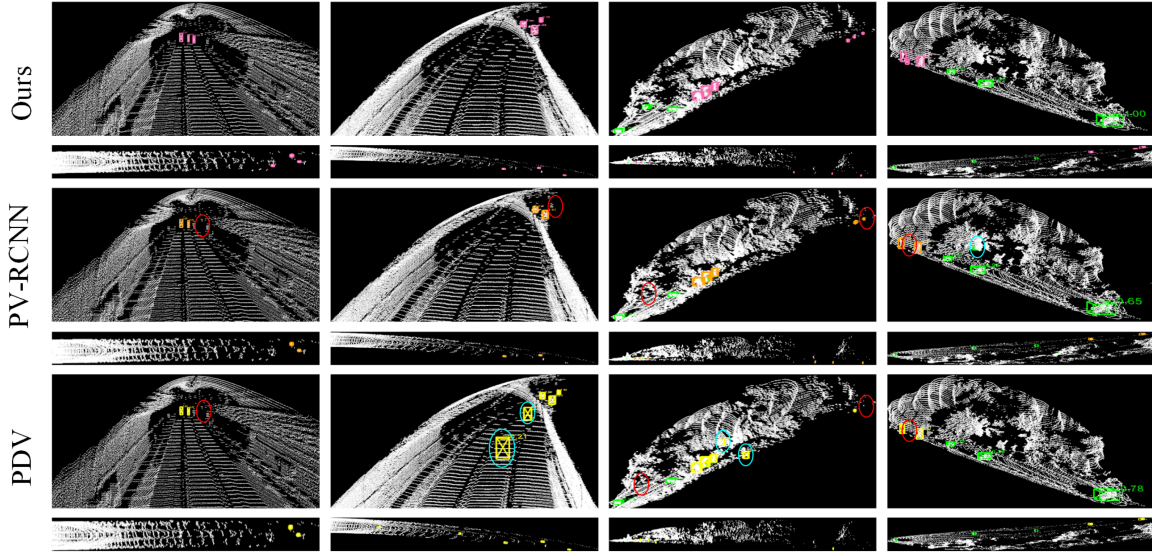


Fig. 5. Visualization of detection results obtained by RAE3D, PV-RCNN, and PDV on the two scenes in the Rail3D validation set. The frontal view and corresponding bird's-eye view of each scene are visualized to facilitate observation of the detection results. Pink, orange, and yellow are used to distinguish pedestrian prediction boxes from different algorithms, while green represents obstacle prediction boxes for all methods. In the front-view point cloud images, red circles highlight missed detections, and cyan circles indicate false positives.

TABLE II

PERFORMANCE COMPARISON ON THE RAIL3D VALIDATION SET AND THE LATENCY IS MEASURED BY US BASED ON THE SAME ENVIRONMENT (I.E., RTX 3090 GPU) WITH 4 BATCH SIZE

Type	Method	Reference	Stage	mAP	Easy	Car Mod.	Hard	Easy	Ped. Mod.	Hard	Easy	Cyc. Mod.	Hard	Latency (ms)
voxel	SECOND	Sensor 2018	one	68.50	90.55	81.61	78.60	55.94	51.15	46.17	82.96	66.74	62.79	50
	PointPillars	CVPR 2019	one	66.60	86.74	78.38	75.17	57.30	51.45	46.87	81.57	62.93	58.96	24
	Part-A2	TPAMI 2020	two	73.95	92.15	82.91	82.05	66.80	59.67	54.56	90.35	70.09	66.97	59
	Voxel R-CNN	AAAI 2021	two	-	92.14	84.98	82.46	-	-	-	-	-	-	74
	SIEV-Net	2022	one	72.89	88.91	78.25	76.39	70.43	66.52	61.16	85.90	66.48	61.95	45
	OcTr	CVPR 2023	one	70.79	88.43	78.57	77.16	61.49	57.17	52.35	85.29	70.44	66.17	64
	MsSVT++	2024	one	74.32	89.24	84.93	78.90	66.37	59.58	53.92	92.63	73.98	69.31	97
Point	PointRCNN	CVPR 2019	two	71.28	91.13	80.60	78.14	61.00	53.86	47.13	91.69	71.17	66.77	100
	3DSSD	CVPR2020	one	70.58	88.55	79.45	77.30	58.91	56.50	50.01	86.79	71.83	65.92	38
Voxel +Point	PV-RCNN	CVPR 2020	two	72.21	92.10	84.36	82.48	62.71	54.50	49.88	88.92	69.19	65.74	98
	SA-SSD	CVPR 2020	one	70.84	90.15	79.91	78.78	63.66	54.62	50.64	85.07	70.16	64.55	40
	PIPC-3Ddet	2024	two	77.81	89.85	86.44	79.31	76.55	70.05	63.93	91.75	72.59	69.81	56
	RAE3D(Ours)	-	one	76.30	91.68	80.31	78.10	72.36	66.40	60.07	91.92	75.32	70.58	35

TABLE III

ABLATION STUDY OF RAE3D COMPONENTS ON KITTI AND RAIL3D VALIDATION SPLITS; THE BASELINE METHOD IS SECOND_IOU, AND THE PERFORMANCE IS EVALUATED USING 3D AP(%) CALCULATED AT 40 RECALL POSITIONS

Datasets	Components	MVA-PQ BEV-MLA L_centroid		✓		✓		✓		✓	
KITTI val	Car	Easy	91.55	91.58	(+0.03)	91.57	(+0.02)	91.62	(+0.07)	91.68	(+0.13)
		Mod.	79.37	79.43	(+0.06)	79.41	(+0.04)	79.9	(+0.53)	80.31	(+0.94)
		Hard	77.62	77.98	(+0.36)	77.88	(+0.26)	78.02	(+0.40)	78.1	(+0.48)
	Ped.	Easy	61.13	69.67	(+8.54)	66.58	(+5.45)	71.36	(+10.23)	72.36	(+11.23)
		Mod.	54.55	62.58	(+8.03)	59.48	(+4.93)	64.98	(+10.43)	66.4	(+11.85)
		Hard	49.48	57.63	(+8.15)	55.5	(+6.02)	59.37	(+9.89)	60.07	(+10.59)
	Cyc.	Easy	90.69	91.61	(+0.92)	91.32	(+0.63)	91.87	(+1.18)	91.92	(+1.23)
		Mod.	71.25	73.26	(+2.01)	72.67	(+1.42)	74.62	(+3.37)	75.32	(+4.07)
		Hard	66.27	68.38	(+2.11)	67.64	(+1.37)	69.85	(+3.58)	70.58	(+4.31)
	3D mAP		71.32	74.68	(+3.36)	73.56	(+2.24)	75.73	(+4.41)	76.30	(+4.98)
Rail3D val	Obs.	3D	68.33	75.53	(+7.20)	72.28	(+3.95)	78.31	(+9.98)	79.36	(+11.03)
	Ped.	3D	89.38	93.13	(+3.75)	91.45	(+2.07)	94.98	(+5.60)	95.65	(+6.27)
	3D mAP		78.86	84.33	(+5.47)	81.87	(+3.01)	86.65	(+7.79)	87.51	(+8.65)

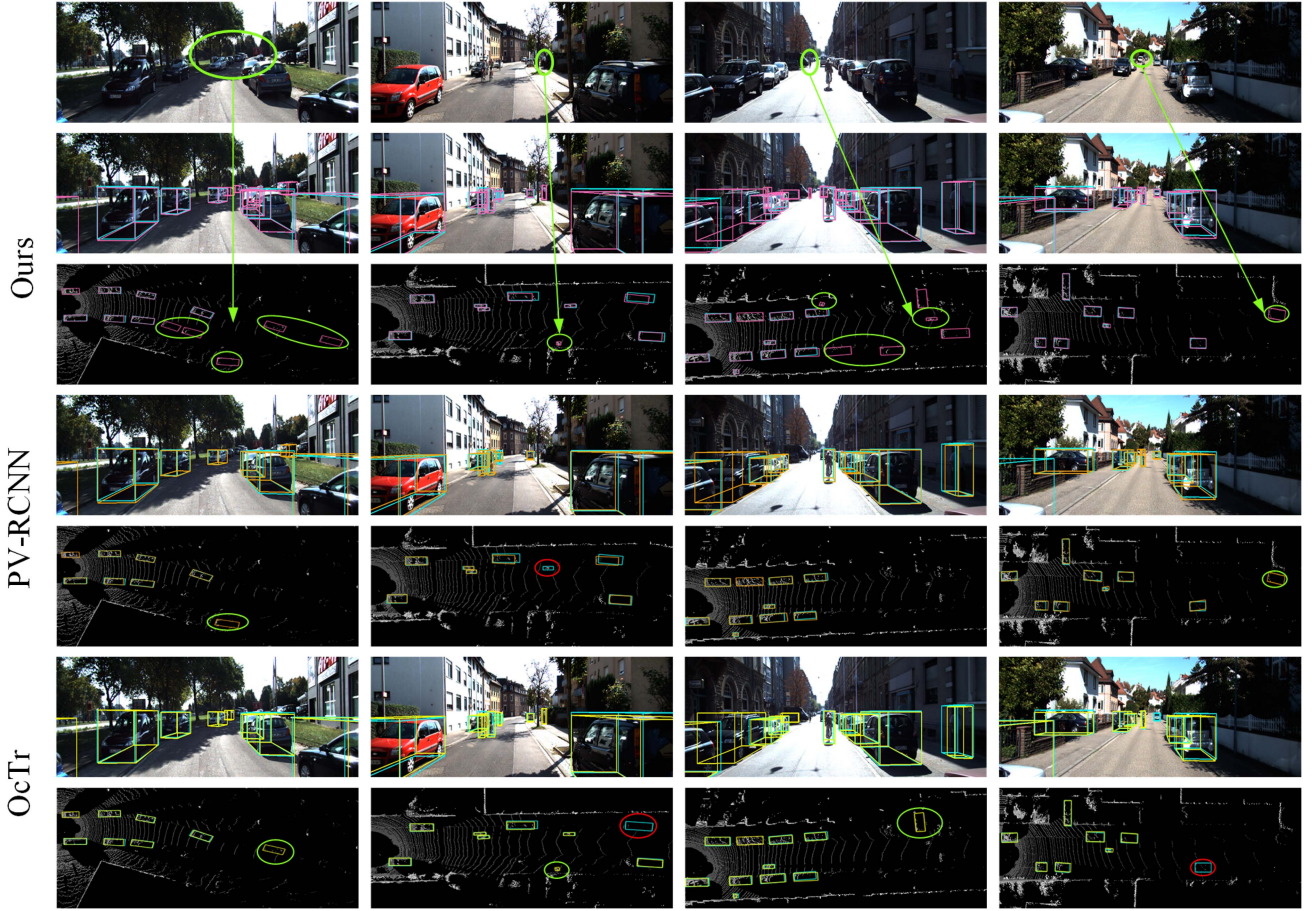


Fig. 6. Visualization of detection results obtained by RAE3D, PV-RCNN, and OcTr in the KITTI validation set. The bird's-eye view of the point cloud and the corresponding RGB image for each scene are visualized to examine the detection results. Pink, orange, and yellow boxes represent predictions from different algorithms, while cyan indicates ground truth boxes. In the point cloud bird's-eye view, red circles highlight missed detections, and green circles mark correct predictions that outperform the ground truth.

Rail3D validation sets, respectively. Similarly, incorporating the BEV-MLA module yields a 2.24% and 3.01% improvement on the same datasets. Both modules provide notable gains for small object detection, demonstrating that the MVA-PQ module enhances the geometric representation of voxel features for small objects through multiscale feature encoding, while the BEV-MLA module enriches the spatial semantic representation of BEV features by integrating shallow voxel information. Importantly, the introduction of the 3D centroids offset loss function, in conjunction with the two modules, allows the proposed model of RAE3D to achieve optimal results, with improvements of 4.98% and 8.65% in 3D mAP on the KITTI and Rail3D validation sets. These findings highlight the individual and combined contributions of the components of RAE3D in advancing 3D object detection performance.

2) *Hyperparameter Searching*: In the overall loss function (13), the parameters λ_1 , λ_2 , λ_3 , and λ_4 serve as weights to balance the multitask loss components. To efficiently determine the optimal weight configuration, we conduct a hierarchical parameter search experiment on the validation set of the Rail3D dataset. The evaluation metric used is the 3D mAP for the “Obstacle” and “Pedestrian” categories.

TABLE IV
PARAMETER SEARCH FOR THE LOSS FUNCTIONS; IN THE BROAD-RANGE EXPLORATION PHASE OF WEIGHT DISTRIBUTION (STAGE-1) AND THE FINE-TUNING PHASE WITHIN THE OPTIMAL RANGE (STAGE-2) ARE EVALUATED BY 3D MAP (%)

Stage-1 (1,2,3,4)	3D mAP	Stage-2 (1,2,3,4)	3D mAP
(1.0, 2.0, 0.2, 0.1)	78.17	(1.0, 2.0, 0.2, 0.8)	78.25
(1.0, 2.0, 0.2, 0.5)	77.87	(1.0, 2.0, 0.2, 1.2)	76.90
(1.0, 2.0, 0.2, 1.0)	78.80	(1.0, 2.0, 0.2, 1.4)	77.11
(1.0, 2.0, 0.2, 2.0)	78.49	(1.0, 2.0, 0.2, 1.6)	78.66
(1.0, 2.0, 0.2, 5.0)	77.53	(1.0, 2.0, 0.3, 1.0)	77.07
(1.0, 1.0, 0.2, 1.0)	78.57	(1.0, 2.0, 0.4, 1.0)	78.78
(1.0, 3.0, 0.2, 1.0)	77.81	(1.0, 2.0, 0.5, 1.0)	78.56
(1.0, 2.0, 0.1, 1.0)	76.94	(1.0, 2.0, 0.3, 0.8)	78.70
(1.0, 2.0, 0.5, 1.0)	78.56	(1.0, 2.0, 0.3, 1.2)	78.47
(1.0, 2.0, 0.2, 0.3)	77.38	(1.0, 2.0, 0.3, 1.4)	78.30

Specifically, following the methodology in [11], the initial values of λ_1 , λ_2 , and λ_3 are set to 1.0, 2.0, and 0.2, respectively. In the first stage, a wide-range parameter search is conducted around these initial loss weights, adjusting λ_2 and λ_3 , while rapidly exploring the range of λ_4 . As shown in Table IV, based on the results of the first stage, a fine-grained search is performed

TABLE V

IMPACT OF THE NUMBER OF 3D SPARSE CONVOLUTION DOWN-SAMPLING OPERATIONS IN THE MVA-PQ MODULE ON DETECTION ACCURACY AND EFFICIENCY (EVALUATED ON THE RAIL3D VALIDATION SET, TRAINED FOR 10 EPOCHS)

3DSPCConv num.	Down-sampling	Obs.	Ped.	3D mAP	Latency (ms)	Param. (M)
N=3	4x	68.61	87.34	77.98	27.26	18.1
N=4	8x	70.35	90.51	80.43	23.52	18.1
N=5	16x	64.32	87.86	76.09	22.07	18.1
N=6	32x	65.51	85.1	75.31	20.89	18.1
N=7	48x	61.43	83.23	72.33	20.25	18.1

within the optimal range during the second stage. The detection performance of RAE3D is optimal when the weights are set to $\lambda_1 = 1.0$, $\lambda_2 = 2.0$, $\lambda_3 = 0.2$, and $\lambda_4 = 1.0$.

3) *Down-Sampling Depth Analysis*: To analyze the impact of the number of down-sampling operations in the MVA-PQ module on detection performance, experiments with varying down-sampling times were conducted. As shown in Table V, “ $N = 3$ ” indicates that the 3D sparse convolution module performs three down-sampling operations on voxel features, and so on. The results show that the proposed method achieves the best detection performance when N is set to 4, with a 3D mAP of 80.43%, obstacle detection accuracy of 70.35%, pedestrian detection accuracy of 90.51%, and latency of 23.52 ms. As N increases (e.g., $N \geq 5$), detection efficiency improves, but accuracy declines, with the 3D mAP dropping to 72.33% at $N = 7$ due to insufficient feature resolution for capturing target details. Conversely, when N is smaller (e.g., $N = 3$), feature retention improves, but computational complexity and latency increase significantly (e.g., 27.26 ms), resulting in sub-optimal performance. Therefore, $N = 4$ is selected to achieve the best balance between accuracy and efficiency in the MVA-PQ module.

V. CONCLUSION

We proposed RAE3D, a single-stage, end-to-end 3D detection architecture specifically optimized for rail transit obstacle perception. To efficiently processed 3D sparse data, the MVA-PQ module combined the global characteristics of 3D sparse convolution with the local precision of SA. This design addressed challenges such as small object detection and point cloud quantization loss in railway scenarios. The BEV-MLA module employed sparse height compression and multiscale BEV feature fusion, significantly enhancing feature representation, receptive field, and real-time performance while handling uneven sparse data distribution. To improve detection accuracy, a 3D centroid offset loss was introduced into the loss function, constraining the geometric centroid offset between predicted and ground-truth bounding boxes. Extensive experiments on KITTI and Rail3D datasets demonstrated that RAE3D achieves state-of-the-art 3D detection performance with superior real-time efficiency. Ablation studies confirmed that each module contributed to performance gains, and their integration enabled RAE3D to achieve optimal results. This architecture was particularly suitable for railway scenarios, where its ability to detect long-range, sparse objects with efficiency and stability was critical.

However, our method still faces challenges in detecting small objects and handling complex scenes. In the future, we plan to explore multimodal fusion approaches to enhance accuracy and robustness in 3D detection performance.

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